Mihai Anitescu

Computational mathematics, optimization, and uncertainty quantification

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HONORS AND AWARDS

- Winner, 2023 INFORMS Computing Society COIN-OR Cup, for the best open-source software for optimization, for ExaModels.jl and MadNLP.jl (with Sungho Shin and Francois Pacaud).
- Winner, INFORMS Best Paper in Energy Award, 2023, for A. Subramanyam, J. Kim, M. Anitescu and others, "Failure probability constrained AC optimal power flow", IEEE Transactions on Power Systems, 37 (2022), pp. 4683-4695.
- Impact Argonne Award, 2021.
- Argonne Physical Sciences and Engineering (PSE) Excellence Award, 2020. With Charlotte Haley, Ray Osborn, and Stephan Rosenkranz
- SIAM Fellow, 2019.
- Winner, IEEE PES Best Conference Paper Award 2019 (4 winners out of 1300 submissions, with Adrian Maldonado and Michel Schanen)
- Featured, DOE's Quadrennial Technology Review, 2015. ("Improving the Energy Grid", Page 352).
- Winner, Best 2013 COAP paper award, (with Miles Lubin, Julian Hall, and Cosmin Petra)
- Finalist, Best Paper Award for the IEEE PES Summer Meeting 2014 (with S. Abhyankar and V. Rao)
- Winner of the 2013 COIN-OR cup (with Miles Lubin, Julian Hall, and Cosmin Petra)
- Recipient of the 2012 DOE INCITE Award (with a 14M CPU hour allocation on the ANL BG/L "Intrepid" Supercomputer)
- Recipient of the 2011 DOE INCITE Award (with a 10M CPU hour allocation on the ANL BG/L "Intrepid" Supercomputer)
- Wilkinson Fellow, Argonne National Laboratory, 1997-1999.
- Silver medal at the International Mathematical Olympiad, 1986.
- Prizes at the regional and national phases of The Physics Competition (both the general and the electromagnetics sections) for Romanian college students, 1989.
- Romanian National Fellowship, 1989.
- First prize at the Romanian National Mathematical Competition "Traian Lalescu" (college students), 1988.
- Multiple prizes at the regional and national phases of the Romanian National Olympiad and various other national high school mathematics and physics competitions, 1984–1986.

FUNDING

- **At Argonne External**
  - co-Investigator and thrust lead, "Energy Efficient Computing: A Holistic Methodology", DOE-ASCR Base, $15M, from FY25.
  - Lead PI "Optimal Design of Carbon Capture Units under Uncertainties in Feed Compositions and Operational Conditions", DOE-FECM through HPC4EI, $400K plus $100K cost share from Shell, 2025-2026 (option for a second phase ongoing).
  - Lead PI "Macroreliability of the U.S. Power Grid: Operational Reliability of Low-Inertia Systems", DOE-OE Advanced Grid Modeling, $1,300,000, from FY24.
  - Site PI "Structural Signatures of Hidden Order in Spin-Orbit Coupled Systems", DOE-BES, $750K, from FY24.

- PI “AGM Transmission Planning”, DOE-OE Advanced Grid Modeling, $1.1M over three years, from FY23.
- Lead PI “Grid Modernization CASCDE”, DOE Grid Modernization Initiative, $3.6M over three years, from FY23.
- Lead, one-year off-ramp extension of MACSER (MMICCS), DOE-ASCR, $500K, FY23.
- Lead PI for Argonne, “Machine-Learning Accelerated Simulations for Forecasting”, DOE-ASCR, $3.2M total with $2M to Argonne, FY22-FY24.
- Lead PI “Graph-Indexed Optimization (Base)”, DOE-ASCR, $1.67M, FY22-FY24.
- co-PI “Probabilistic Machine Learning for Rapid Large-scale and High-rate Aerostructure Manufacturing (PRISM)”, DOE-EERE Applied Manufacturing, $2.4M, FY22-FY23 (lead Prasanna Balaprakash).
- Supplement to the Exascale ExaSGD project, $200K for Argonne, FY22-FY23.
- One-year extension, “MACRO resilience simulation to power grid”, DOE-OE, $400K.
- PI “Macro-Res of the North Am Pow Grid”, DOE-OE,$375K,FY20-FY22.
- PI “ Accurate and Tractable Cascade Failure Models”, DOE-OE,$390K,FY20-FY21.
- co-PI “Structural Signatures of Hidden Order in Spin-Orbit Coupled Systems”,DOE-BES,$1M, FY20-FY22
- PI “Space-time Indexed Optimization Problems”, DOE-ASCR,$440K, FY19-FY21.
- PI, “Resilience-Constrained Grid Planning”, DOE-OE-AGM, $400K, 2017-2019.
- co-PI, “Bilevel Optimization Approaches for N-k Reliability Computations”, $300K, 2017-2019. (PI, Kibaek Kim),
- co-PI, “Stochastic Optimization for Distribution System Uncertainty”, $375K, 2017-2019 (PI, Shrirang Abhyankar),
- PI and Director, “MACSER: Multifaceted Mathematics for Complex Energy and Environmental Systems”, DOE-ASCR, $ 3M, 2018-2023. At award time, the project had 27 PIs from 6 sites: Argonne, PNNL, LLNL, Univ. of Wisconsin, Univ. of Chicago, Ohio State.
- co-PI and site lead, “ECP: Stochastic Optimization at the Exascale for Planning and Cybersecurity”, $ 2.5M(+1.5M,2019-2022), 2016-2022.
- PI, “Data-Driven Optimization for Complex Dynamical Systems”, 440K, 2016-2019.
- PI, Grid Modernization Laboratory Consortium 1.4.18 (“Computational Science for the Grid”), 1M, 2016-2019.
- PI, “Spatio-Temporal Statistical Analysis at Scales Using Gaussian Processes”, DOE-ASCR, $ 400K, 2013-2016.
- PI and Director, “M2ACS: Multifaceted Mathematics for Complex Energy Systems”, DOE-ASCR, $ 3.5M, 2012-2017. At award time, the project had 22 PIs from 5 sites: Argonne, PNNL, Sandia, Univ. of Wisconsin, Univ. of Chicago.
- PI, “Scalable Dynamic Optimization”, DOE-ASCR, $400K, 2012-2015.
- co-PI “ Frameworks, Algorithms, and Scalable Technologies for Mathematics (FASTMath) SciDAC Institute”, DOE-ASCR, $ 6.75M, 2011-2016. PI: Lori Diachin. ANL PI: Barry Smith. I am the lead PI for Differential Variational Inequalities subtopic, at $250K.
- co-PI “Climate Science for a Sustainable Energy Future (CSSEF)”, DOE-BER, $ 5M 2011-2016; Lead ANL UQ PI. Lead PI: David Bader (ORNL).
- co-PI “Center for Exascale Simulation of Advanced Reactors (CESAR)”, DOE-ASCR,$4M 2011-2016. Lead of the Uncertainty Quantification Area. Lead PI: Robert Rosner (University of Chicago).
- PI, “Scalable statistical analysis of Gaussian models for Petascale spatiotemporal data”, DOE-ASCR, $450K, 2009-2013.
- PI, “Stochastic Optimization of Complex Systems”, DOE-ASCR, $485K, 2009-2012.
- PI, “Advanced Numerical Methods for Differential Variational Inequalities”, $325K, DOE-ASCR, 2009-2012.

- co-PI "SCIDAC-E: Differential Variational Inequalities for Phase Field Problems", DOE-ASCR, $300K, 2009-2011. Lead PI: Lois Curfman McInnes.
- Co-PI, "International Symposium on Mathematical Programming", $20K, DOE-ASCR, 2009.
- PI, " A New Challenge for Computational Science: Complementarity Constraints", $375K, DOE-ASCR, 2006-2009.
- Co-PI, NSF DMS-0937025, "International Symposium on Mathematical Programming", $20K, 2009.

- **At Argonne External on Annual Renewal Schedules**
  - Annual Operation Plan for Officed of Energy Delivery co-PI, "Development of methodologies for large-scale optimization for electrical transmission planning". PI: Victor Zavala, 2015-2016,$300K (and 2010-2011, $375K, 2011-2012, $375K, 2012-2013, $375K, 2013-2014, $375K, 2014-2015, $375K).
  - Workpackage Manager " VV UQ for Reactor IPSC", DOE-NE, $300K, 2011–2012 (and 2010-2011, $200K).
- **At Argonne Internal**
  - Co-Investigator of the LDRD project 2025-0487, "Energy-Efficient Foundation Model for Power Grid Resilience Analysis", $50K, 2025 (PI Feng Qiu).
  - Co-Investigator of the "Big Data Grand Challenge" LDRD Award, $1.5M, 2012-2015.
  - Co-Investigator of the "Mesoscale Electro-Mechanical Materials" LDRD Award, $400K, 2012-2015.
  - Co-Investigator of the "Toward Understanding Cloud Processes and Uncertainty Modeling in Next-Generation High-Resolution Climate Models", LDRD Award, $250K, 2011-2014
  - Co-Investigator of the "Novel Power System Operations Methods for Wind-Powered Systemr" LDRD Award, by Argonne National Laboratory, $350K, 2009-2012
  - Co-Investigator of the "Separations" LDRD Award, by Argonne National Laboratory, $300K, 2007-2010
  - Co-Investigator of the "Virtual Fab Lab" LDRD Award, by Argonne National Laboratory $300K+, 2004-2006 (PIs P. Zapol and S. Gray).
- **At the University of Chicago**
  - FACCTS (France and Chicago Collaborating in the Sciences) Award, "Quantum Control", University of Chicago, $10K, 2025.
  - PI of the proposal "CPS:Synergy:GOALI:Collaborative Research: Real-time Data Analytics for Energy Cyber-Physical Systems". 384K, 2015-2018.
  - co-PI of the proposal "RTG: Computational and Applied Mathematics in Statistical Science". 1.7M, 2016-2021, PI John Lafferty.
- **At Pitt**
  - Co Principal Investigator for the National Science Foundation Grant DMS-0112239 "Scientific Computing Research Environments for the Mathematical Sciences (SCREMS)", 2001-2003, $25,000 + $39,000 University of Pittsburgh cost sharing (with J. Chadam, C. Chow, W. Layton, and I. Yotov).
  - Principal Investigator for the National Science Foundation Grant DMS-9973071, "A Computational Framework for Multi-Rigid Body Dynamics with Contact and Friction", 1999-2002, $75,000.
  - Principal Investigator for the University of Pittsburgh Grant "A Computational Framework for Multi-Rigid Body Dynamics with Contact and Friction", 1999-2001, $14,204.

TEACHING ACTIVITIES

- **University of Pittsburgh**
  - Classes taught: Elementary functions, Calculus for Engineering, Trigonometric Functions, Calculus for Business, Introduction to Linear Programming, Introduction to Numerical Linear Algebra, Introduction to Scientific Computing (graduate), Introduction to Numerical Ordinary Differential Equations (graduate), Industrial Mathematics, Numerical Linear Algebra (graduate), Introduction to Numerical Partial Differential Equations (graduate).
  - *Curriculum Development*

    * Developed the graduate class Introduction to Scientific Computing (1999).
    * Developed, together with John Burkardt of the Pittsburgh Supercomputing Center, the computational laboratory for Introduction to Scientific Computing (1999).
    * Participated in various curriculum enhancement activities of the undergraduate mathematics program, including a Sloan Foundation-sponsored development of an industrial mathematics program (2000).

- **University of Chicago, Graduate Classes**
  - Stats 31015: Computational Mathematics II: Convex Optimization, 2017, 2019
  - Stats 31020: Computational Mathematics II: Nonlinear Programming. 2014, 2016,2018,2020
  - Stats 343: Linear Models. 2012,2013,2014.
  - Stat 365: Sequential Estimation, 2016.
  - Stats 310: Computational Mathematics II: Simulation and Optimization. Winter, 2012, 2011, 2010
- **University of Chicago, Undergraduate Classes**
  - Stats 245: Statistical Theory and Methods II, 2020.
  - Math 211: Intro to Numerical Analysis

MENTORSHIP AND SUPERVISION ACTIVITIES

- **Postdoctoral scholars and long-term visitors, Argonne: 33**
  - Dan Negrut (2004-2005), currently Professor at University of Wisconsin, Department of Mechanical Engineering. Research Topic: " Scalable Multiscale Methods for Orbital-Free Density Functional Theory "
  - Victor Zavala (2008-2010) Research Topic: "Optimization of Hybrid Energy Systems". Currently, Associate Professor, University of Wisconsin
  - Emil Constantinescu (2008-2010). Research Topic: "Uncertainty Quantification". Currently, Computational Mathematician, Argonne National Laboratory
  - Xiaoyan Zeng (2008-2012). Research Topic: "Uncertainty Analysis in Chemical Plant Safety and Climate". Currently, Assistant Professor at Shanghai University
  - Cosmin Petra (2009-2012). Research Topic: "Stochastic Programming". Currently, Computational Mathematician, Lawrence Livermore National Laboratory
  - Oleg Roderick (2009-2012). Research Topic: "Hybrid Sampling-Sensitivity Approaches for Nuclear Reactors". First employment: scientist at Argonne. Currently, Data Scientist at Verizon
  - Lei Wang (2010-2013). Research Topic: "Variational Inequalities in the Simulation of Heterogeneous Materials". Currently, Assistant Professor at University of Wisconsin, Milwaukee
  - Jungho Lee (2010-2013, co-advised with Barry Smith). Research Topic: "Variational Inequalities in the Simulation of Heterogeneous Materials". Currently, at Citibank's Fixed Income group
  - Drew Khouri (2012-2013, Wilkinson Fellow). Research Topic: "Uncertainty Quantification and Optimization. Staff Scientist at Sandia National Laboratory
  - Jie Chen (2010-2014). Research Topic: "Numerical Linear Algebra for Gaussian Process Simulation". Currently, Staff Scientist at IBM
  - Ilias Bilionis (2013-2014). Research Topic. Research Topic: "Spatio-temporal statistical models for Solar and Wind Power". Currently Assistant Professor at Purdue University
  - Linyun Liang (2013-2016). Research Topic. Research Topic: "Phase Field Models". Currently Faculty at Beihang University, Beijing, China
  - Hong Zhang (2014-2016). Research Topic: "Adjoints for Energy Systems". Currently, staff Scientist at Argonne
  - Francois Gilbert (2014-2017). Research Topic:"Scalable Dynamic Optimization". First employment: technical staff at Argonne. Currently: Amazon
  - Zhen Zhang (2014-2015). Research Topic:"Uncertainty Quantification". Currently, Assistant Profes-

sor at UC Santa Barbara
  - Julie Bessac (2014-2017). Research Topic:“Spatio-Temporal Statistics”. (with Emil Constantinescu). Currently, staff Scientist at Argonne
  - Charlotte Haley (2014-2017). Research Topic:“Spatio-Temporal Statistics”. Currently, staff Scientist at Argonne
  - Michel Schanen (2015-2017). Research Topic:“Parallel Methods for Adjoint Computations”. Currently, staff Scientist at Argonne
  - Feng Qiang (2015-2017). Research Topic:“Stochastic Programming”. Currently employment in industry (China)
  - Adrian Maldonado (2017-2019). Research Topic: “Uncertainty Propagation through Dynamic Transients.” Following assignment: Staff scientist at Argonne
  - Brian Dandurand (2017-2020). Research Topic: “Stochastic Programming” (with Kibaek Kim). Following assignment: lecturer at UW-Eau Claire
  - Vishwas Hebbur Vankata Rao (2017-2019). Research Topic: “Data Assimilation” (with Emil Constantinescu). Currently, staff scientist at Argonne
  - Ahmed Attia (2017-2019). Research Topic: “Data Assimilation”. (with Emil Constantinescu). Following assignment: staff scientist at Argonne
  - Youngdae Kim (2018-2022). Research Topic: “Real-time Optimization”. Following assignment: Scientist at Exxon-Mobil
  - Johannes Brust (2018-present). Research Topic: “Structured quasi-Newton methods”. Following assignment: lecturer at UCSD
  - Anirudh Subramanyam (2018-2022). Research Topic: “Optimization under Uncertainty”. Following assignment: tenure-track assistant professor at Penn State OR
  - Amanda Lenzi (2019-present). Research Topic: “Space time statistics”. Following assignment: lecturer, University of Edinburgh
  - Francois Pacaud (2020-2022). Research Topic: “ Condensed Methods for GPU”. Following assignments: faculty, Ecole des Mines, Paris, France
  - Sungho Shin (2021-2024). Research Topic:“ Graph-indexed Optimization”. Following assignment: faculty at MIT
  - Mitchell Krock (2023-2024). Subsequent position: faculty at the University of Missouri
  - Mohammad S. Ramadan (2023-)
  - Philip Dinenis (2024-)
  - Alexis Montoison (2025-)
- **Long-term visiting scholars, Argonne: 1**
  - Alexis Montoison (2023), student visitor
- **Full-time Argonne employees supervised: 15**
  - Emil Constantinescu, Assistant Computational Mathematician, Argonne National Laboratory, 2010-2015. First subsequent position: Computational Mathematician, Argonne
  - Victor Zavala, Assistant Computational Mathematician, Argonne National Laboratory, 2010-2015. First subsequent position: Computational Mathematician, Argonne
  - Miles Lubin, Predoctoral Appointee, Argonne National Laboratory, 2011-2012 (first subsequent position: Ph.D Student at MIT)
  - Cosmin Petra, Assistant Computational Mathematician, 2012-2016. First subsequent position: Computational Mathematician, LLNL
  - Oleg Roderick, Assistant Computational Mathematician, 2012-2014. Currently, Data Scientist at Geisinger Health Systems
  - Jie Chen, Assistant Computational Mathematician, 2014-2014. First subsequent position: IBM
  - Francois Gilbert, Mathematics Specialist, 2017-2018. OR Researcher, Amazon
  - Kibaek Kim, Assistant Computational Mathematician, 2016-2022

- Michel Schanen, Computer Science Specialist, 2017-
- Charlotte Haley, Statistician, 2017-
- Julie Bessac, Statistician, 2017-2018. First subsequent position, Statistician, Argonne
- Vishwas Rao, Computational Mathematician, 2019-
- Adrian Maldonado, Energy Systems Scientist, 2019-
- Paul Beckman, Assistant Computational Mathematician, 2019-2020 (first subsequent position, Ph.D student at NYU)
- Shrirang Abhyankar, Argonne National Laboratory, 2014

- **Predoctoral appointees, Argonne: 5**
  - Chris Geoga, Predoctoral Appointee, 2016-2018
  - Chris Geoga, Student Appointee, 2018-2022 (first subsequent position: Ph.D student at Rutgers)
  - Jacob Roth, Predoctoral Appointee, 2018-2020 (first subsequent psoition currently Ph.D at U of Minn)
  - Albert Lam, Predoctoral Appointee, 2020-2022 (first subsequent position: Moderna)
  - Miao Li, Predoctoral Appointee, 2022-
- **Student interns, Argonne: 42**
  - Alec Hanson (Princeton, with Lois C. McInnes), 1998. Project: "Parallel Implementation of Time Dependent Differential Variational Inequalities"
  - Adrian Dunca (University of Pittsburgh, with Traian Iliescu), 2002. Project: " Optimal Design of Fluid Flow Using Subproblems Reduced by Large Eddy Simulation "
  - Jufeng Peng (Rensselaer) 2003. Project: "Mathematical Programs with Complementarity Constraints in Robotics". Co-author of 1 conference proceedings paper (ICRA 2003)
  - Bogdan Gavrea (University of Maryland, Baltimore County), 2005. Project: "Quadratic Programming Approaches for Multi-Body Dynamics with Contact and Friction". Co-author on 2 journal submissions
  - Gun Srijutongsiri (grad, Cornell), 2005. Project: "Statistics of Granular Flow"
  - Adrian Kopacz (Northwestern), 2005. Project: " Scalable Methods for Orbital-Free Density Functional Theory Calculations"
  - Monika Neda (University of Pittsburgh), 2006. Project: "A collocation approach for uncertainty quantification in nuclear reactors". Co-author of 1 conference proceedings paper (ANS M&S 2007)
  - Xiaoyan Zeng (Illinois Institute of Technology), 2006. Project: "Chemical Plant Safety Assessment Under Uncertainty" Co-author of 1 journal paper
  - Emil Constantinescu (Virginia Tech), 2006. Project: " Scalable Methods for Orbital-Free Density Functional Theory Calculations" Co-author of 1 conference proceedings papers
  - Oleg Roderick (Portland State University), 2008-2009. Project: " A Hybrid Sampling Sensitivity Approach for Uncertainty Quantification in Nuclear Reactors " co-author of 2 journal papers and 1 conference proceedings paper
  - Kyle Schmitt, (MIT), 2008. Project: " Efficient Sampling of Dynamical Systems with Spatial Uncertainty " co-author of 1 conference proceedings paper and 1 journal paper
  - Matt Rockhlin, (University of Chicago, with Emil Constantinescu), 2009. Project: "Uncertainty Quantification of Numerical Weather Prediction Systems". Co-author of 1 journal paper
  - Sangmin Lee (NYU, with Victor Zavala), 2009. Project: "Stochastic Unit Commitment under Uncertainty". Co-author of 1 journal paper
  - Brian Lockwood (University of Wyoming, CSGF fellow), 2010. Project: " Gaussian Process Models with Derivative Information". Co-author of 1 journal paper
  - Zhu Wang (Virginia Tech, with Oleg Roderick), 2010. Project " Reduced Order Modeling in Nuclear Reactor Applications". Co-author of 1 journal paper
  - Miles Lubin (University of Chicago, with Cosmin Petra), 2010-2011. Project "Parallel Software for Stochastic Programming". Co-author of 1 journal paper and 1 conference proceedings paper

- Toby Heyn (University of Wisconsin, with Dan Negrut), 2010. Project “Iterative Solvers for Differential Variational Inequalities”. Co-author of 1 proceedings paper
- Hayes Stripling (Texas A&M University), 2011. Project “ Uncertainty Analysis of Wave Reactors”. Co-author of 1 journal paper
- Zhu Wang (Virginia Tech, with Oleg Roderick), 2011. Project “ Using POD in Nuclear Reactor Applications”
- Carmeline D’Silva (Princeton), 2012. Project “Uncertainty Analysis of Molecular Dynamics Simulations”
- Wanting Xu (Chicago). 2014, 2015 Project " Low Memory Methods for weak 4DVar
- Ahmed Attia (Virginia Tech). 2014 Project “Optimization methods for importance sampling”
- Jing Yu (Chicago). 2015, 2016 “Design of Experiments for Gas Pipelines”
- Vivak Patel (Chicago). 2016: “Inversion of Dynamic Parameters of Grid”
- Shan Lu (Chicago). 2016: “Demand Models for Planning Problems”
- Richard Chen (Chicago), 2017: “Structured Linear Algebra for Gaussian Process Calculations”
- Youngseok Kim (Chicago), 2017: “SQP methods for nonparametric Bayesian estimation”
- Vivak Patel (Chicago), 2017: “Inversion of Solar Power Deployment Factors”
- Youngseok Kim (Chicago), 2018: “SQP methods for nonparametric Bayesian estimation”
- Sen Na (Chicago), 2018: “Sensitivity of Dynamic Optimization”
- Rui Liu (Georgia Tech), 2018: “Network Optimization under Uncertainty”
- Sungho Shin (Wisconsin) 2018: “Decentralized Control”
- Sen Na (Chicago), 2019: “Temporal Decomposition of Nonconvex Dynamic Programming”
- Yian Chen (Chicago), 2019: “Implicit Gaussian Processes”
- Jess Kunke (Chicago), 2019: “space time statistical models for solar power”
- Sen Na (Chicago), 2020: “Online Nonconvex Optimization”
- Yian Chen (Chicago), 2020: “HODLR approaches to statistical modeling”
- Mihai Alexe, 2010. Summer student at Argonne.
- Justin Madsen, 2008. Summer student at Argonne.
- Hammad Mazhar, 2009. Summer student at Argonne.
- Arman Pazouki. Summer student at Argonne.
- Matthew Rocklin, 2011. Summer student at Argonne.

- **Ph.D. students advised, University of Chicago: 6**
  - Wanting Xu, UChicago (with M. Stein; Ph.D 2017; first employment Morgan Stanley)
  - Vivak Patel, UChicago (Ph.D 2018, first employment: tenure-track Assistant Professor at the University of Wisconsin, Madison)
  - Jing Yu, UChicago, (Ph.D 2019, first employment: Citibank)
  - Xialiang Dou, UChicago (PhD 2022, first employment: Two Sigma)
  - Sen Na, Uchicago (PhD 2021, first employment, UC Berkeley)
  - Yian Chen, Uchicago (PhD 2023)
- **M.S. students advised, University of Chicago: 10**
  - Yuxuan Ren, 2021, MS
  - Jess Kunke, 2020, MS
  - Albert Lam, 2020, MS
  - Jisung Hwang, 2020, MS
  - Jacob Roth, 2018, MS
  - Shan Lu, 2017, MS
  - Yuchen Su, 2017, MS
  - Tom Hen, 2017, MS

- David Francis McDiarmid 2013, MS
- Miles Lubin 2011, MS

- **Undergraduate theses advised, University of Chicago: 2**
  - Katie Zimmerman, 2017
  - Paul Beckman, 2019
- **Ph.D. students advised, University of Pittsburgh: 2**
  - Gary D. Hart, Ph.D., 2007. (with William J. Layton). Currently Instructor, University of Pittsburgh at Greensburgh. Ph.D: "A Constrained-Stabilized Time-Stepping Approach for Piecewise Smooth Multibody Dynamics. "
  - Faranak Pahlevani, Ph.D., 2004 (with William J. Layton). Ph.D Thesis: " Sensitivity Analysis of Eddy Viscosity Models". Currently Assistant Professor at Penn State Abington
- **Degree and examination committees: 3**
  - Comprehensive Exam Committee (at Pitt 1999-2002) Students: Atife Caglar, Jon Drover, Adrian Dunca, Noel Heitmann, Faranak Pahlevani, and Niyazi Sahin, all Math Department at Pitt
  - MS Committess Mark Fenner, Department of Computer Science, University of Pittsburgh (2002)
  - Ph.D Committees Anasthasos Liakos, Niyazi Sahin, Adrian Dunca, Hussein-Al Attas, Kimberley Jordan (all at the University of Pittsburgh mathematics department, 1999-2002), Pierre Dognin (electrical engineering, University of Pittsburgh, 2003), Arvind Pupil Raghunathan (chemical engineering, Carnegie Mellon University, 2004), Bogdan Gavrea (mathematics, University of Maryland, Baltimore County, 2006) Cosmin Petra (mathematics, University of Maryland, Baltimore County, 2009), Brian Lockwood (Wyoming, Mechanical Engineering, 2011), Joe Guinness, (U Chicago, Stats, 2012), Toby Heyn (Wisconsin, Mechanical Engineering, 2013); Hayes Stripling (Texas A &M, Nuclear Engineering, 2013); Jouke deBaar (Delft, 2014); Somak Dutta, U Chicago Stats, 2015; Michael Horell, U Chicago Stats, 2015; Jonathan Eskreis-Winkler, 2019; Yongrui Chen, 2020; Youngseok Kim, 2020;Haoyang Liu, 2020

REFEREEING AND EDITORIAL SERVICE

- **Editorial Boards**
  - Associate Editor, Journal of Optimization Theory and Applications (since 2025).
  - Senior Editor, Optimization Methods and Software (since 2007; Software Editor 2004-2007).
  - Associate Editor, SIAM Journal on Optimization (since 2010).
  - Associate Editor, SIAM/ASA Journal on Uncertainty Quantification (2012-2019).
  - Associate Editor, Mathematical Programming, series A (2004-2017).
  - Associate Editor, Mathematical Programming, series B (2007-2017).
  - Associate Editor, SIAM Journal on Scientific Computing (2011-2016).
- **National Academy Report Review**
  - Analytic Research Foundations for the Next-Generation Electric Grid, 2016.
- **Proposal Reviews**
  - Simons Foundation Pivot Fellowship (2025).
  - The Department of Energy Early Career program (2021, 2022, 2023).
  - Kaust Research Fund (2018)
  - National Science Foundation. (1998, 1999, 2002, 2006, 2009),
  - Romanian Science Foundation (2005).
  - The Natural Science and Engineering Research Council of Canada (2004),
  - The Hong Kong Science Foundation (2002).
  - The Department of Energy (2008,2009,2010,2011,2012, 2013,2014, 2015, 2016, 2017,2018,2019).
- **Panel participation**

  - NSF Optimization panel (2024).
  - NSF Focused Research Groups (FRG) panel (2024).
  - NSF-ITR small grants panel (2002),
  - NSF Operations Research panel (2006),
  - NSF CSUMS panel (2006).
  - DOE Early Career Awards Panel (2009).
  - NSF Operation Research Panel (2015).
  - NSF Tripods Panel (2017).
  - NSF Computational Math Early Career Panel (2018)
- **Invited NSF workshop participation**
  - "Benchmarks for High Performance Computing", (2005).
  - "Mathematics in Robotics", (2000).
- **Other**
  - Refereed papers (about 15-20 a year) for SIAM Journal in Optimization, SIAM Journal Of Control, SIAM Journal Of Numerical Analysis, SIAM Journal On Scientific Computing, Numerische Mathematik, Mathematical Programming, Applied Mathematics Letters, Linear Algebra and Its Applications, Computational Optimization and Applications, Optimization Methods and Software, Control, Optimization, and the Calculus of Variations, Optimization and Engineering, Journal of Optimization Theory and Applications, ACM Transactions on Graphics, ACM SIGGRAPH, International Journal of Numerical Methods in Engineering, IEEE Transactions in Signal Processing, IEEE Transactions in Robotics, IEEE International Conference in Robotics and Automation, Transactions of the Institute of Industrial Engineers, Computers and Chemical Engineering.

PROFESSIONAL SOCIETY SERVICE

- **Professional Society Officer**
  - SAMSI Director Search Committee, 2016-2017.
  - SIAM representative to the Statistical and Applied Mathematical Sciences Institute (SAMSI) Governing Board. 1/1/15-12/31/20
  - Vice-president, the SIAM Interest group in Optimization (2011-2013).
  - Vice-chair for linear programming and complementarity, Optimization Section, Institute for Operations Research and Management Science (INFORMS), 2005-2008. Organized the Linear Programming/ Complementarity Clusters at INFORMS annual meeting, 2006-2007.
- **Conference and Workshops Organization Committees**
  - ICERM Rare Events workshop, 2019.
  - IMSI Long Program "Uncertainty Quantification and AI for Complex Systems", March 3 - May 23, 2025 (organizing committee).
  - ICCOPT 2022, co-organizer of the applications track (with Christian Kirches).
  - IMSI Workshop "Expressing and Exploiting Structure in Modeling, Theory, and Computation with Gaussian Processes", 2022, co-organized with Daniel Sanz-Alonso and Michael Stein.
  - IMSI Workshop on Data Reduction in Genomics, August-September 2021 (hybrid), co-organized with M. Stephens, D. Nicolae and A. Gilbert.
  - IMSI Workshop on Verification, Validation and Uncertainty Quantification, May 2021 (online), co-organized with C. Graziani, F. Cattaneo and R. Rosner.
  - MACSER annual meeting, February 2021 (online; project lead).
  - ACNW (Argonne-Chicago-Northwestern-Wisconsin) Optimization and Machine Learning Workshop. June, 2017.
  - SIAM UQ Conferences, Lausanne, 2016
  - Optimization and UQ Workshop, IMA, Minneapolis, February 22-February 24, 2016.

- Applied Mathematics PI meeting, Albuquerque, 2013.
- Supercomputing 2013, member of the technical program committee.
- Supercomputing 2015, member of the technical program committee.
- MOS International Conference on Continuous Optimization, 2013, Portugal, Application Cluster co-organizer.
- The 2nd International Workshop on High Performance Computing, Networking and Analytics for the Power Grid, held in conjunction with Supercomputing 2012, Nov 2012, Salt Lake City, UT.
- The 1st International Workshop on High Performance Computing, Networking and Analytics for the Power Grid, held in conjunction with Supercomputing 2011, Nov 2011, Seattle.
- The DOE "Mathematics for the Analysis, Simulation, and Optimization of Complex Systems" Workshop, Sep 2011, Crystal City.
- Verification, Validation and Uncertainty Quantification Across Disciplines, 2011, Park City Utah (a workshop fully sponsored by DOE – participants got all travel and local expenses funded – through the Institute for Computation in Science).
- Optimization in Energy Systems, 2010, Snowbird Utah (a workshop fully sponsored by DOE through the Institute for Computation in Science).
- The DOE Crosscut Extreme Scale Workshop, 2010, Rockville, MD.
- The 2nd INL/NCSU Verification and Validation Workshop, Myrtle Beach, SC, 2010.
- ICCOPT 2010: The 3rd International Conference on Continuous Optimization of the Mathematical Programming Society, Santiago de Chile.
- SIAM Annual Meeting, 2009, Denver
- International Symposium for Mathematical Programming, 2009, Chicago.
- Steering Committee, Midwest Numerical Analysis Conference, (2007-present).
- The 15-th, 16-th, 17-th, 18-th International Conference on Control Systems and Computer Science, Bucharest, Romania, 2005, 2007, 2009, 2011 (sponsored by the regional IEEE chapter).
- Joint EUROPT-OMS Meeting 2007: 2nd Conference on Optimization Methods and Software and 6th EUROPT Workshop on Advances in Continuous Optimization July 4-7, 2007, in Prague, Czech Republic (member of program committee).
- World Congress in Computational Mechanics, 2006, Los Angeles (member of advisory committee)

- **Minisymposia organized**
  - Characterization and Prediction of Rare and Extreme Events in Complex Systems, SIAM Conference on Computational Science and Engineering, 2021 (two-part minisymposium).
  - Optimization INFORMS fall meeting, San Antonio, 2000; INFORMS fall meeting, San Jose 2002 (2); International Symposium in Mathematical Programming, Copenhagen, 2003; INFORMS fall meeting, Atlanta 2003 (2); Canadian Operations Research Society international meeting, Banff, 2004; INFORMS fall meeting, Denver 2004 (3), International Conference on Complementarity Problems, Stanford 2005; INFORMS fall meeting San Francisco 2005 (2); 2-nd International Conference on Continuous Optimization, Hamilton, Canada, 2007; INFORMS Annual Meeting 2007, Seattle; INFORMS Annual Meeting Washington DC, 2008; INFORMS Annual Meeting 2009, San Diego (2); SIAM CSE Conference (2011, 2); SIAM Parallel Processing Conference, 2012; International Symposium in Mathematical Programming, 2012 (3), Berlin; SIAM CSE Conference, Boston, 2013, SIAM UQ Conference, Savannah 2014, SIAM Optimization Conference, San Diego 2014, SIAM Annual Meeting Chicago 2014; SIAM UQ Conference; SIAM Optimization 2017, Vancouver (3); SIAM MPE Philadelphia 2018; SIAM CSE Spokane 2019; Mathematical Optimization of Systems Impacted by Rare, High-Impact Random Events, ICERM 2019, Providence RI,
  - Multibody Dynamics World Congress in Computational Mechanics, Los Angeles, 2006; International Conference for Intelligent Robots and Systems, 2003 (Dynamics Section).
  - Uncertainty Quantification Idaho Verification and Validation Workshop, Myrtle Beach, (2010), SIAM UQ Conference, Raleigh, 2012;
  - Artificial Intelligence Energy-Fusion Energy Sciences Breakout at the ANL 2019 Artificial Intelligence

Meeting (with Bill Tang); Smart Energy Infrastructure track at the DOE Town Hall Meeting in AI, Wash DC 2019 (with Teja Kuruganti, Mihai Anitescu, Mary Ann Piette, Tianzhen Hong).

PRIZE AND SPECIAL LECTURE COMMITTEE WORK

- OMS Broyden Prize Committee 2022 (Chair)
- OMS Broyden Prize Committee 2021.
- SIAM UQ Officer Nomination Committee, 2022.
- SIAM SIAG/OPT Best Paper Prize Committee, 2020 (Chair)
- Mathematical Optimization Society – Society of Industrial and Applied Mathematics (MOS-SIAM) 2015 Lagrange Prize Committee (chair).
- International Congress of Mathematicians ICM 2014, Control and Optimization Panel.
- Society for Industrial and Applied Mathematics (SIAM) Optimization Activity Group Prize Committee, 2013 (chair).
- INFORMS Coin-OR Cup Prize Committee, 2014.
- INFORMS Coin-OR Cup Prize Committee, 2025 (Chair).
- OMS Broyden Best Paper Prize Committee, 2024-2025.
- SIAM UQ Early Career Prize Committee, 2021.

COMMITTEE WORK

- **University of Chicago**
  - Computational and Applied Mathematics Graduate Admission Committee, 2018-
  - Computational and Applied Mathematics Undergraduate Major Steering Committee 2014-.
  - Computational and Applied Mathematics Initiative Hiring Committee, 2012-2014.
- **Argonne National Laboratory**
  - MCS (divisional) strategic council (2018-)
  - Co-chaired the Data Forum (2012-2014)
  - Chaired the Division Director Search Committee (2010-2011).
  - Director's Postdoctoral Committee (2010-2012).
  - Chaired the Wilkinson Selection Committee (2012, 2010, 2008).
  - Divisional Awards Committee (2009,2010).
- **University of Pittsburgh**
  - Served in the Mellon Chair Search Committee (1999).
  - Departmental Chair Selection Committee (2000).
  - Scientific Computing Hiring Committee (2002).

INVITED TALKS

- **2021-2025**
  - "Optimization for Digital Twins of Weather and Power", New York Academy of Sciences panel, September 2025
  - "Scalability and Sustainability Tools for Digital Twins from Power Grid Applications", Colloquium, Flatiron Institute, September 2025
  - "Scalable Multi-Period ACOPF utilizing GPUs with High Memory Capacities", International Conference on Continuous Optimization, Los Angeles, July 2025
  - "Exponential Decay of Sensitivity in Dynamic and other Graph-Indexed Programming and Computational Consequences", USC Operations Research Colloquium, April 2025

- “Tail dependence as a measure of teleconnected warm and cold extremes of North American wintertime temperatures”, SIAM Conference on Computational Science and Engineering, 2025
- “Reliability, Extreme Weather Events, and Downscaling for Energy Applications”, Notre Dame Statistics Colloquium, March 2025
- “Reliability, Extreme Weather Events, and Downscaling for Energy Applications”, NYU Tandon School of Engineering Workshop on Decision under Uncertainty for Energy, January 2025
- “Issues in Modeling and Computing with Extremes”, NSF Workshop on Data-driven Modeling and Prediction of Rare and Extreme Events, Institute for Mathematical and Statistical Innovation, November 2024
- “GPU-accelerated nonlinear programming”, INFORMS Annual Meeting, Seattle, October 2024
- “Condensed Space Methods for Nonlinear Programming on GPU”, International Symposium on Mathematical Programming, Montreal, July 2024
- “Large Deviation Approaches for Resilience Calculations”, IMSI Green Energy Workshop, July 2024
- “Condensed Space Methods for Nonlinear Programming on GPU”, NREL 7th Autonomous Systems Workshop, 2024
- “The Two Language Problem; Productivity in Technical Computing; Domain Specific Languages; Rapid Digital Twins”, University of Chicago Statistics Seminar, December 2023
- “Exponential Decay of Sensitivity in Graph-Indexed Optimization”, IIT seminar, September 2023
- “Rapid Digital Twins in Power Grid”, National Academies Symposium on Digital Twins, July 2023 (virtual)
- “Scalable Interior Point Algorithms on GPU”, FERC software conference, June 2023
- “Exponential Decay of Sensitivity in Graph-Indexed Optimization”, SIAM Conference on Optimization, May 2023
- “Rare Events Computation in Power Grid and Climate”, Clemson Extreme Event workshop, May 2023
- “Exponential Decay of Sensitivity in Graph-Indexed Optimization”, 12th US-Mexico Workshop on Optimization and its Applications, January 2023
- “Scalable Gaussian Process Calculations”, RWTH Aachen, June 2022 (remote seminar)
- “Scalable Gaussian Process Calculations”, ORNL AIRES Workshop, April 2022 (remote)
- “Rare event simulation of energy cascades using Kinetic Monte Carlo”, IEEE PES Society-Wide seminar, October 2021
- “Exponential Decay of Sensitivity in Dynamic Programming and Computational Consequences”, US Naval Academy Conference in Optimization, June 2021
- “Rare event simulation and control of electricity cascades”, PHILMS virtual seminar, April 2021
- “Rare event simulation and control of electricity cascades”, SIAM Conference on Computational Science and Engineering, 2021 (virtual)

- **2016-2020**
  - IMA Seminar on Industrial Mathematics (2016)
  - AMS Meeting, Denver, (2016)
  - FERC Software Conference (Washington, DC, June 2016)
  - SAMSI Optimization Workshop, Research Triangle Park, August 2016
  - SIAM CSE, Atlanta (2017)
  - SIAM Optimization Conference (2017)
  - Applied Math Colloquium at U of Iowa (2017)
  - Computational Science and Engineering Seminar (Notre Dame, 2018)
  - The 9th annual graduate student mini-conference in Computational Mathematics, Columbia, SC (2018, keynote)
  - 7th International Conference on High Performance Scientific Computing, Hanoi, Vietnam (2018, plenary)

  - Purdue Data Science Day, Purdue, IN (2018)
  - SIAM UQ Conference, Orange Grove, CA (invited, 2018)
  - IMA Workshop on Forecasting for Complex Processes (invited, 2018)
  - The International Symposium on Mathematical Programming, Bordeaux, France (invited, 2018)
  - INFORMS annual meeting 2018 (invited)
  - SIAM Conference on Mathematics of Planet Earth, 2018 (invited panelist)
  - IEEE PES 2019 (invited panelist by the UW working group)
  - Computing in Engineering Forum 2019, Wisconsin Madison (invited)
  - Computing in Engineering Forum 2020, (virtual, invited)
  - IEEE PES HPC Working Group Webinar, 2020 (invited, virtual)
  - IEEE PES Webinar, 2020 (invited).
- **2011-2015**
  - SIAM Computational Science and Engineering Conference, Reno (2011)
  - SIAM Optimization Meeting, Darmstadt, Germany, (2011)
  - International Conference in Control Systems and Computer Science, Bucharest, Romania (plenary, 2011)
  - IMA Workshop on UQ, Minneapolis, (2011)
  - SAMSI Summer School on UQ, Albuquerque (invited lecturer, 2011)
  - The DOE Applied Math Research PI Meeting 2011, (plenary)
  - University of Texas, ICES, 2011 (Seminar)
  - University of Pittsburgh 2012
  - SIAM UQ Conference 2012
  - University of Chicago 2012
  - the International Symposium on Mathematical Programming, Berlin, 2012
  - University of Illinois (the IMSE kick-off conference), 2012
  - DTRA CyberPhysical Networks Workshop, Washington DC, 2012
  - International Conference on Complementarity Problems, Singapore, 2012
  - Powertech 2013, Grenoble, 2013
  - ACCES Workshop, Aachen, 2013
  - SIAM Optimization Conference, San Diego, 2014
  - SIAM UQ Conference, Savannah 2014
  - International Conference on Complementarity Problems, Berlin, 2014
  - ACNW Workshop, Chicago 2015
  - NREL, Golden, 2015
  - IMSP Pittsburgh 2015
  - INFORMS, Philadelphia 2015 (keynote)
  - EPRI Innovation Summit, Huntington Beach, October 2015
  - ABB Colloquium, Raleigh, December 2015
- **2006-2010**
  - Illinois Institute of Technology (2006)
  - Workshop on Assessment of Sensitivity/Uncertainty Analysis Capabilities Applicable for the Nuclear Fuel Cycle, North Carolina State University (2006)
  - 3-rd Workshop on the Global Nuclear Energy partnership, Washington, D.C. (2006)
  - International Congress on Industrial and Applied Mathematics, Zurich (2007)
  - International Conference on Continuous Optimization, Hamilton, (semi-plenary, 2007)
  - Computational and Mathematical Methods in Science and Engineering, (plenary, 2007)
  - International Workshop on Hybrid Systems Modeling, Simulation and Optimization Koc University,

Istanbul (2008)
  - GNEP Verification and Validation workshop, Idaho Falls (2008)
  - Stevens Institute of Technology Colloquium (2008)
  - INFORMS Annual Meeting, Washington DC (2008)
  - SIAM Conference on Computational Science and Engineering, Miami (2009)
  - University of Illinois at Chicago Math Colloquium (2009)
  - American Nuclear Society Annual Meeting, Atlanta (2009, roundtable keynote)
  - International Symposium on Mathematical Programming, Chicago (2009, semi-plenary)
  - SIAM Annual Meeting, Denver, (2009)
  - US National Congress in Computational Mechanics, Columbus, (2009)
  - INFORMS Annual Meeting (2009)
  - Lehigh High Performance Computing Meeting (2009, plenary)
  - SIAM Parallel Processing Conference, Seattle (2010)
  - ANS Annual Meeting San Diego (2010)
  - 2nd Verification and Validation Workshop, Myrtle Beach (2010)
  - 12-th International Conference on Stochastic Programming, Halifax, (2010)
  - Princeton University, Operations Research and Financial Engineering Colloquium (2010)
  - University of Wisconsin, Computer Science Colloquium (2010)
- **2001-2005**
  - Old Dominion University (2001)
  - IMA Workshop on Haptics, Virtual Reality and Human Computer Interaction, Minneapolis (2001)
  - Contact Mechanics International Symposium, Peniche, Portugal, (2001)
  - International Conference on Scientific Computation And Differential Equations, Vancouver, Canada (2001)
  - INFORMS fall meeting, Miami Beach (2001)
  - SIAM Optimization Conference, Toronto, (2002)
  - International Symposium on Complementarity Problems, Cambridge, (2002)
  - INFORMS fall meeting, San Jose, (2002)
  - International Congress on Computational and Applied Mathematics, Sydney (2003)
  - SIAM Annual Meeting, Montreal (2003)
  - INFORMS fall meeting, Atlanta (2003)
  - International Conference on Intelligent Robots and Systems, Las Vegas (2003)
  - SIAM Annual Meeting, Portland (2004)
  - Sandia Multiscale Optimization Workshop (2004)
  - BIRS Workshop on Molecular Dynamics, Banff (2005)
  - Midwest Numerical Analysis Conference, (plenary, 2005)
  - 15th International Conference on Control and Computer Science, Bucharest, (2005)
  - 4th International Conference on Complementarity Problems, Stanford, (2005)
- **1997-2000**
  - Georgia Institute of Technology (1997)
  - Clemson University (1997)
  - Argonne National Laboratory (1997)
  - University of Louisville (1997)
  - University of Pittsburgh (1997)
  - University of Iowa (1997)
  - SIAM Conference in Optimization, Atlanta (1999)

– INFORMS fall meeting, Philadelphia (1999)
– University of Maryland, Baltimore County (1999)
– INFORMS spring meeting, Salt Lake City (2000)
– Mathematical Programming Symposium, Atlanta (2000)
– INFORMS fall meeting, San Antonio (2000)

PUBLICATIONS

JOURNAL PAPERS PUBLISHED OR IN PRINT

1. Pacaud, François, Sungho Shin, Alexis Montoison, Michel Schanen, and Mihai Anitescu. "Condensed interior-point methods for scalable nonlinear programming on GPUs." Mathematical Programming Computation (2026). doi:10.1007/s12532-026-00335-0.

2. Ramadan, Mohammad S., Evan Toler, and Mihai Anitescu. "Data-Conforming Data-Driven Control: Avoiding Premature Generalizations Beyond Data." IEEE Transactions on Automatic Control (2026). doi:10.1109/tac.2025.3647312.

3. Shin, Sungho, Sen Na, and Mihai Anitescu. "Near-Optimal Performance of Stochastic Model Predictive Control." Mathematics of Operations Research (2026). doi:10.1287/moor.2023.0159.

4. Kuang, Wei, Mihai Anitescu, and Sen Na. "Online covariance matrix estimation in sketched Newton methods." Information and Inference: A Journal of the IMA (2026). doi:10.1093/imaiai/iaag012.

5. Dobson, Ian, D. Adrian Maldonado, and Mihai Anitescu. "The Statistical Spread of Transmission Outages on a Fast Protection Time Scale Based on Utility Data." IEEE Transactions on Power Systems (2026). doi:10.1109/tpwrs.2025.3596141.

6. Rossmann, Ramsey, Mihai Anitescu, Julie Bessac, Michael Ferris, Mitchell Krock, James Luedtke, and Line Roald. "A framework for balancing power grid efficiency and risk with bi-objective stochastic integer optimization." Optimization and Engineering (2025). doi:10.1007/s11081-024-09944-x.

7. Kim, Youngdae, François Pacaud, Michel Schanen, Kibaek Kim, and Mihai Anitescu. "Leveraging GPU Batching for Scalable Nonlinear Programming through Massive Lagrangian Decomposition." SIAM Journal on Scientific Computing (2025). doi:10.1137/21m1450112.

8. Anderson, Edward, Michael Ferris, Andrew Philpott, Mihai Anitescu, Peter Cramton, Sijia Geng, Richard Green, Tito Homem-de-Mello, Olivier Huber, Vincent Leclère, and Ramteen Sioshansi. "Ten challenges for mathematical modeling of the green-energy transition." Current Sustainable/Renewable Energy Reports (2025). doi:10.1007/s40518-025-00274-9.

9. Na, Sen, Mihai Anitescu, and Mladen Kolar. "A Fast Temporal Decomposition Procedure for Long-Horizon Nonlinear Dynamic Programming." Mathematics of Operations Research (2024). doi:10.1287/moor.2023.1378.

10. Shin, Sungho, Mihai Anitescu, and François Pacaud. "Accelerating optimal power flow with GPUs: SIMD abstraction of nonlinear programs and condensed-space interior-point methods." Electric Power Systems Research (2024). doi:10.1016/j.epsr.2024.110651.

11. Bhaduri, Anindya, Nesar Ramachandra, Sandipp Krishnan Ravi, Lele Luan, Piyush Pandita, Prasanna Balaprakash, Mihai Anitescu, Changjie Sun, and Liping Wang. "Efficient Mapping Between Void Shapes and Stress Fields Using Deep Convolutional Neural Networks With Sparse Data." Journal of Computing and Information Science in Engineering (2024). doi:10.1115/1.4064622.

12. Ramadan, Mohammad S., and Mihai Anitescu. "Extended Kalman Filter–Koopman Operator for Tractable Stochastic Optimal Control." IEEE Control Systems Letters (2024). doi:10.1109/lcsys.2024.3410889.

13. Pacaud, François, Michel Schanen, Sungho Shin, Daniel Adrian Maldonado, and Mihai Anitescu. "Parallel interior-point solver for block-structured nonlinear programs on SIMD/GPU architectures." Optimization Methods and Software (2024). doi:10.1080/10556788.2024.2329646.

14. Shin, Sungho, Mihai Anitescu, and Victor M. Zavala. "Exponential decay of sensitivity in graph-structured nonlinear programs." SIAM Journal on Optimization 32.2 (2022): 1156-1183. Awarded the 2023 W. David Smith, Jr. Graduate Student Paper Award given by the Computing and Systems Technology Division of the AIChE; to Sungho Shin.

15. Pacaud, François, Sungho Shin, Michel Schanen, Daniel Adrian Maldonado, and Mihai Anitescu. "Accelerating Condensed Interior-Point Methods on SIMD/GPU Architectures." Journal of Optimization Theory and Applications (2023). doi:10.1007/s10957-022-02129-5.

16. Lenzi, Amanda, and Mihai Anitescu. "How can statistics help to prevent blackouts?." Significance (2023). doi:10.1093/jrssig/qmad009.

17. Na, Sen, Mihai Anitescu, and Mladen Kolar. "Inequality constrained stochastic nonlinear optimization via active-set sequential quadratic programming." Mathematical Programming (2023). doi:10.1007/s10107-023-01935-7.

18. Shin, Sungho, Yiheng Lin, Guannan Qu, Adam Wierman, and Mihai Anitescu. "Near-Optimal Distributed Linear-Quadratic Regulator for Networked Systems." SIAM Journal on Control and Optimization (2023). doi:10.1137/22m1489836.

19. Chen, Yian, and Mihai Anitescu. "Scalable Physics-Based Maximum Likelihood Estimation Using Hierarchical Matrices." SIAM/ASA Journal on Uncertainty Quantification (2023). doi:10.1137/21m1458880.

20. Beckman, Paul G., Christopher J. Geoga, Michael L. Stein, and Mihai Anitescu. "Scalable computations for nonstationary Gaussian processes." Statistics and Computing (2023). doi:10.1007/s11222-023-10252-0.

21. Na, Sen, and Mihai Anitescu. "Superconvergence of Online Optimization for Model Predictive Control." IEEE Transactions on Automatic Control (2023). doi:10.1109/tac.2022.3223323.

22. Pacaud, François, Daniel Adrian Maldonado, Sungho Shin, Michel Schanen, and Mihai Anitescu. "A feasible reduced space method for real-time optimal power flow." Electric Power Systems Research (2022). doi:10.1016/j.epsr.2022.108268.

23. Na, Sen, Mihai Anitescu, and Mladen Kolar. "An adaptive stochastic sequential quadratic programming with differentiable exact augmented lagrangians." Mathematical Programming (2022). doi:10.1007/s10107-022-01846-z.

24. Brust, Johannes J., and Mihai Anitescu. "Convergence Analysis of Fixed Point Chance Constrained Optimal Power Flow Problems." IEEE Transactions on Power Systems (2022). doi:10.1109/tpwrs.2022.3146873.

25. Lenzi, Amanda, Julie Bessac, and Mihai Anitescu. "Detecting Large Frequency Excursions in the Power Grid With Bayesian Decision Theory." IEEE Open Access Journal of Power and Energy (2022). doi:10.1109/oajpe.2022.3144612.

26. Tan, Bendong, Junbo Zhao, Nan Duan, Daniel Adrian Maldonado, Yingchen Zhang, Hongming Zhang, and Mihai Anitescu. "Distributed Frequency Divider for Power System Bus Frequency Online Estimation Considering Virtual Inertia From DFIGs." IEEE Journal on Emerging and Selected Topics in Circuits and Systems (2022). doi:10.1109/jetcas.2022.3141975.

27. Subramanyam, Anirudh, Jacob Roth, Albert Lam, and Mihai Anitescu. "Failure Probability Constrained AC Optimal Power Flow." IEEE Transactions on Power Systems (2022). doi:10.1109/tpwrs.2022.3146377.

28. Lenzi, Amanda, and Mihai Anitescu. "How Statistics Keeps the Lights on." Significance (2022). doi:10.1111/1740-9713.01704.

29. Rao, Vishwas, Charlotte Haley, and Mihai Anitescu. "LaplaceInterpolation.jl: A Julia package for fast interpolation on a grid." Journal of Open Source Software (2022). doi:10.21105/joss.03766.

30. Na, Sen, Sungho Shin, Mihai Anitescu, and Victor M. Zavala. "On the Convergence of Overlapping Schwarz Decomposition for Nonlinear Optimal Control." IEEE Transactions on Automatic Control (2022). doi:10.1109/tac.2022.3194087.

31. Upreti, Puspa, Matthew Krogstad, Charlotte Haley, Mihai Anitescu, Vishwas Rao, Lekh Poudel, Omar Chmaissem, Stephan Rosenkranz, and Raymond Osborn. “Order-Disorder Transitions in (CaxSr1-x)3Rh4Sn13.” Physical Review Letters (2022). doi:10.1103/physrevlett.128.095701.

32. Maldonado, Daniel Adrian, Emil Mihai Constantinescu, Hong Zhang, Vishwas Rao, and Mihai Anitescu. “Trust-Region Approximation of Extreme Trajectories in Power System Dynamics.” IEEE Transactions on Power Systems (2022). doi:10.1109/tpwrs.2022.3141372.

33. Schanen, M, V Churavy, Y Kim, and M Anitescu. “Rapid Prototyping with Julia: From Mathematics to Fast Code.” Collections 55 (06) (2022).

34. Rao, Vishwas, and Mihai Anitescu. “Efficient Computation of Extreme Excursion Probabilities for Dynamical Systems through Rice’s Formula.” SIAM/ASA Journal on Uncertainty Quantification (2021). doi:10.1137/19m1308177.

35. Lenzi, Amanda, Julie Bessac, and Mihai Anitescu. “Power grid frequency prediction using spatiotemporal modeling.” Statistical Analysis and Data Mining: The ASA Data Science Journal (2021). doi:10.1002/sam.11535.

36. Chen, Yian, and Mihai Anitescu. “SCALABLE GAUSSIAN PROCESS ANALYSIS FOR IMPLICIT PHYSICS-BASED COVARIANCE MODELS.” International Journal for Uncertainty Quantification (2021). doi:10.1615/int.j.uncertaintyquantification.2021036657.

37. Geoga, Christopher J., Mihai Anitescu, and Michael L. Stein. “Flexible nonstationary spatio-temporal modeling of high-frequency monitoring data”. To appear in Environmetrics. arXiv preprint arXiv:2007.11418 (2020).

38. Na, Sen, and Mihai Anitescu. “Exponential decay in the sensitivity analysis of nonlinear dynamic programming.” SIAM Journal on Optimization 30, no. 2 (2020): 1527-1554.

39. Shin, Sungho, Victor M. Zavala, and Mihai Anitescu. “Decentralized schemes with overlap for solving graph-structured optimization problems.” IEEE Transactions on Control of Network Systems 7, no. 3 (2020): 1225-1236.

40. Geoga, Christopher J., Mihai Anitescu, and Michael L. Stein. “Scalable Gaussian process computations using hierarchical matrices.” Journal of Computational and Graphical Statistics 29, no. 2 (2020): 227-237.

41. Kim, Youngseok, Peter Carbonetto, Matthew Stephens, and Mihai Anitescu. “A fast algorithm for maximum likelihood estimation of mixture proportions using sequential quadratic programming.” Journal of Computational and Graphical Statistics 29, no. 2 (2020): 261-273.

42. Schanen, Michel, Daniel Adrian Maldonado, François Pacaud, Alexis Montoison, Mihai Anitescu, Kibaek Kim, Youngdae Kim, Vishwas Rao, and Anirudh Subramanyam. “Julia as a portable high-level language for numerical solvers of power flow equations on GPU architectures.” 2020. doi:10.13140/rg.2.2.22427.39205.

43. Roth, Jacob, David A. Barajas-Solano, Panos Stinis, Jonathan Weare, and Mihai Anitescu. “A Kinetic Monte Carlo Approach for Simulating Cascading Transmission Line Failure.” To appear in SIAM Journal on Multiscale Modeling and Simulation. arXiv preprint arXiv:1912.08081 (2019).

44. Xu, Wanting, and Mihai Anitescu. “Exponentially convergent receding horizon strategy for constrained optimal control.” Vietnam Journal of Mathematics 47, no. 4 (2019): 897-929.

45. Yu, Jing, and Mihai Anitescu. “Multidimensional sum-up rounding for integer programming in optimal experimental design.” Mathematical Programming (2019): 1-40.

46. Dou, Xialiang, and Mihai Anitescu. “Distributionally robust optimization with correlated data from vector autoregressive processes.” Operations Research Letters 47, no. 4 (2019): 294-299.

47. Petra, Cosmin G., Naiyuan Chiang, and Mihai Anitescu. “A structured quasi-newton algorithm for optimizing with incomplete hessian information.” SIAM Journal on Optimization 29, no. 2 (2019): 1048-1075.

48. Xu, Wanting, and Mihai Anitescu. “Exponentially accurate temporal decomposition for long-horizon linear-quadratic dynamic optimization.” SIAM Journal on Optimization 28, no. 3 (2018): 2541-2573.

49. Geoga, Christopher J., Charlotte L. Haley, Andrew R. Siegel, and Mihai Anitescu. “Frequency–wavenumber spectral analysis of spatio-temporal flows.” Journal of Fluid Mechanics 848 (2018).

50. Linyun Liang, Zhi-Gang Mei, Yeon Soo Kim, Mihai Anitescu, and Abdellatif M. Yacou Three-dimensional phase-field simulations of intragranular gas bubble evolution in irradiated U-Mo fuel. Computational Materials Science, Volume 145, 2018, Pages 86-95.

51. Julie Bessac, Emil M. Constantinescu and Mihai Anitescu. “Stochastic simulation of predictive space-time scenarios of wind speed using observations and physical models, ”, “Annals of Applied Statistics”, Volume 12, Number 1 (March 2018), 432-458.

52. Yu, Jing, Victor M. Zavala, and Mihai Anitescu. “A scalable design of experiments framework for optimal sensor placement.” Journal of Process Control, Volume 67, Pages 44-55, July 2018.

53. C. L. Haley and M. Anitescu, “Optimal Bandwidth for Multitaper Spectrum Estimation,” in IEEE Signal Processing Letters, vol. 24, no. 11, pp. 1696-1700, Nov. 2017.

54. Abhyankar, Shrirang; Geng, Guangchao; Anitescu, Mihai; Wang, Xiaoyu; Dinavahi, Venkata: ’Solution techniques for transient stability-constrained optimal power flow Part I, IET Generation, Transmission &; Distribution, 2017, 11, (12), p. 3177-3185.

55. Zhang, Hong, Shrirang Abhyankar, Emil Constantinescu, and Mihai Anitescu. “Discrete Adjoint Sensitivity Analysis of Hybrid Dynamical Systems With Switching.” IEEE Transactions on Circuits and Systems I: Regular Papers 64, no. 5 (2017): 1247-1259.

56. Petra, Noemi, Cosmin G. Petra, Zheng Zhang, Emil M. Constantinescu, and Mihai Anitescu. “A Bayesian approach for parameter estimation with uncertainty for dynamic power systems.” IEEE Transactions on Power Systems 32, no. 4 (2017): 2735-2743. Preprint ANL/MCS-P5525-0116.

57. Anitescu, Mihai, Jie Chen, and Michael L. Stein. “An inversion-free estimating equations approach for gaussian process models.” Journal of Computational and Graphical Statistics 26.1 (2017): 98-107.

58. Zavala, Victor M., Kibaek Kim, Mihai Anitescu, and John Birge. “A stochastic electricity market clearing formulation with consistent pricing properties.” Operations Research 65, no. 3 (2017): 557-576.

59. Linyun Liang, Zhi-Gang Mei,, Yeon Soo Kim, Bei Ye, Gerard Hofman, Mihai Anitescu, Abdellatif M. Yacout. “Mesoscale model for fission-induced recrystallization in U-7Mo alloy” Computational Materials Science Volume 124, November 2016, Pages 228-237.

60. Xu, Wanting, and Mihai Anitescu. “A Limited-Memory Multiple Shooting Method for Weakly Constrained Variational Data Assimilation.” SIAM Journal on Numerical Analysis 54, no. 6 (2016): 3300-3331

61. Petra, Cosmin G., Victor M. Zavala, Elias D. Nino-Ruiz, and Mihai Anitescu. “A high-performance computing framework for analyzing the economic impacts of wind correlation.” Electric Power Systems Research 141 (2016): 372-380.

62. Zhi-Gang Mei*, Linyun Liang, Yeon Soo Kim, Tom Wiencek, Edward O’Hare, Abdellatif M. Yacout, Gerard Hofman, Mihai Anitescu. Grain growth in U-7Mo alloy: a combined first-principles and phase field study. Journal of Nuclear Materials Volume 473, May 2016, Pages 300

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